10	**(i)**		$\overrightarrow{AC} = \begin{pmatrix} 4-b \\ 2 \end{pmatrix}, \overrightarrow{AB} = \begin{pmatrix} 2-b \\ 0 \end{pmatrix}$	**M1**	**1.1**	Forming vectors for sides AB and AC	Implied by next M1
			$(4-a)^2 + (2-b)^2 = (2-a)^2 + (4-b)^2 + 4$ o.e.	**M1**	**1.1a**	Use of AB = AC	
			$16 - 8a + a^2 + 4 - 4b + b^2 = 4 - 4a + a^2 + 16 - 8b + b^2 + 4$	**M1**	**1.1**	expanding	
			$4a - 4b + 4 = 0 \Rightarrow a - b + 1 = 0$	**A1**	**2.1**	**AG** Convincing completion	
				[4]			

Question	Answer	Marks	AOs	Guidance	
10 (ii)	D has position vector $\begin{pmatrix}3\\3\\1\end{pmatrix}$ where D is midpoint of BC	**B1**	**3.1a**	Midpoint OR if clearly minimising AC or AB - **M1** for relevant vector using a and b (May be implied by second M1)	
	$\overrightarrow{AD} = \begin{pmatrix}3-a\\2-a\\1\end{pmatrix}$	**M1**	**1.1**	Finding relevant vector in terms of a or b only	
	Area $= \frac{1}{2}AD.BC = \frac{2\sqrt{3}\sqrt{(3-a)^2+(2-a)^2+1}}{2}$	**M1**	**1.1**	Expression for AD or AD^2 (correct method but may have errors)	May use area proportional to AD, AC or AB without calculation of expression for area
	$\sqrt{3}\sqrt{2\left((a-2.5)^2+0.75\right)}$	**M1**	**3.1a**	Completion of square	Or differentiation of AD, AD^2, AC, AB, AC^2 or AB^2.
	$a = 2.5$ for min	**A1**	**2.2a**		
	Position vector $\begin{pmatrix}2.5\\3.5\\0\end{pmatrix}$	**A1**	**3.2a**		
		[6]			